# An Efficient New Methodology for the Synthesis of 1-Functionalized 2-Halo-2-alkenes via Hydrohalogenation Reaction of Electron-Deficient Allenes

Shengming Ma,* Lintao Li

State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, Shanghai 200032, P. R. China

Fax +86-21-64166128; E-mail: masm@pub.sioc.ac.cn

*Received 4 April 2001*

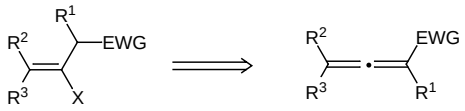
**Abstract:** The hydrohalogenation reactions of electron-deficient allenes, such as 1,2-allenic ketones, carboxylic acids, esters, amides, nitriles, sulfoxides (with the use of $AlX_3/H_2O$), sulfones, and phosphine oxides with MX (X = I, Br, Cl, M = Na, Li) in different solvents, such as HOAc, $CF_3COOH$-HOAc (1:1), $CF_3COOH$ and THF were studied. The reaction is nucleophilic conjugate addition in nature and provides efficient entries to 1-functionalized 2-halo-2-alkenes. The β,γ- over α,β-selectivity and stereoselectivity depend on the solvent(s) and the temperature applied. In most cases, the *Z*- and *E*-isomers can be easily separated with chromatography on silica gel. The subsequent cross coupling reaction with organometallic reagents and terminal alkynes showed the synthetic potential of these products.

1 Introduction
2 Hydrohalogenation Reaction of 1,2-Allenic Ketones
3 Hydrohalogenation Reaction of 1,2-Allenic Carboxylic Acids
4 Hydrohalogenation Reaction of 1,2-Allenic Carboxylates
5 Hydrohalogenation Reaction of 2,3-Dienamides
6 Hydrohalogenation Reaction of 2,3-Dienenitriles
7 Hydrohalogenation Reaction of 1,2-Allenic Sulfones
8 Hydrohalogenation Reaction of 1,2-Allenic Sulfoxides
9 Hydrohalogenation Reaction of 1,2-Allenic Diphenylphosphine Oxides
10 Synthetic Application
11 Conclusion

**Key words:** electron-deficient allenes, hydrohalogenation, regioselectivity, stereoselectivity, solvent effect

## 1 Introduction

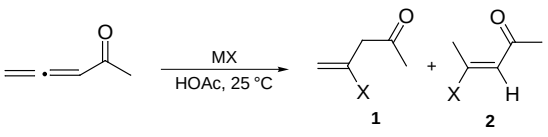
1-Functionalized 2-halo-2-alkenes are classes of compounds with synthetic importance due to the uniquely located carbon-carbon double bonds. Allenes have been found to have high and unique reactivities, due to the existence of the two orthogonal π-bonds.[1] During the course of our study on the chemistry of allene,[2] we rationalized from the retrosynthetic analysis that the hydrohalogenation reaction of electron-deficient allenes would be one of the most efficient ways for the synthesis of β-halogen substituted β,γ−unsaturated products (equation 1). Since the hydrohalogenation of electron-deficient alkynes afforded 1-functionalized 2-halo-1-alkenes,[3] a challenge encountered here would be the potential migration of the C=C bond from the β,γ-position to the α,β-position, thus, affording a synthetically useless mixture of β,γ−unsaturated products and α,β−unsaturated products.[4] We have developed the hydrohalogenation reaction of a varieties of electron-deficient allenes, such as 1,2-allenic ketones, carboxylic acids, esters, amides, nitriles, sulfoxides (with the use of $AlX_3/H_2O$), sulfones, and phosphine oxides with MX in HOAc, $CF_3COOH$-HOAc, $CF_3COOH$ or THF to afford 1-functionalized 2-halo-2-alkenes. Here, the reaction medium is the key to avoid the formation of a mixture of β,γ−unsaturated and α,β−unsaturated products.




**Equation 1**

## 2 Hydrohalogenation Reaction of 1,2-Allenic Ketones

Hydrochlorination reaction of 1,2-allenyl ketones by HCl was reported to lead to polymerization, while hydrochlorination using *N*,*N*'-dimethylhydrazine dihydrochloride or $SnCl_4$ afforded a mixture of β,γ-unsaturated and α,β-unsaturated β-chloroenones, majoring the α,β-enones.[5] We found that reaction of 3,4-pentadien-2-one with NaI, LiI, LiBr•$H_2O$, and LiCl•$H_2O$ in acetic acid at 25 °C afforded the corresponding 4-halo-4-penten-2-one with yields ranging from 69% to 82% (equation 2).[6] Under different reaction conditions, the reaction of 3,4-pentadien-2-one with LiI or NaI afforded different ratio of β,γ-unsaturated enone **1** / α,β-unsaturated enone **2**. Reaction of 3,4-pentadien-2-one with NaI in acetic acid at 25 °C afforded the mixture of β,γ-unsaturated enone **1** and α,β-unsaturated enone **2** with a ratio of 91:9. The reaction can also occur in a 1:1 mixture of HOAc and THF, but the ratio of β,γ-unsaturated enone **1** / α,β-unsaturated enone **2** is 87:13, while in a solution of HOAc (3 equiv.) in THF at a lower reaction temperature the selectivity of the reaction




**Equation 2**



Downloaded by: University of Pittsburgh. Copyrighted material.

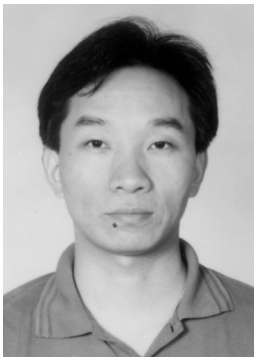
is much higher. The best results were obtained using MI (M = Na, Li), HOAc (3 equiv.) in THF at 0 °C and the ratio of β,γ-unsaturated enone **1** / α,β-unsaturated enone **2** is 98:2.[7]

### Typical Procedure for a Larger Scale Preparation

*Synthesis of 4-Bromo-3-penten-2-one*:[8] A three-necked, round-bottomed flask equipped with a magnetic stirring bar, an addition funnel, a thermometer, and a reflux condenser was flushed with nitrogen and charged with 132 g (0.5 mol) of triphenylphosphine and 400 mL of dichloromethane. The flask was cooled with an ice-water bath, and a solution of 80 g (0.5 mol) of bromine in 50 mL of dichloromethane was added dropwise so that the temperature of the mixture did not rise above 10 °C. After the addition was over, a solution of 50 g (0.5 mol) of acetylacetone in 50 mL of dichloromethane was added dropwise within 10 min at room temperature. The mixture was then heated under reflux for 4-5 h with stirring until the evolution of hydrogen bromide ceased. The cooled mixture was concentrated using a rotary evaporator. 500 mL of ether was added to the residue and triphenylphosphine oxide was precipitated. The resulting precipitate was removed by suction and washed with ether (2×100 mL). The filtrate and washings were combined and evaporated. Distillation of the residue under reduced pressure gave 70 g (85%) of 4-bromo-3-penten-2-one; bp 54-55 °C/15 mmHg.

*Synthesis of 3,4-Pentadien-2-one*:[8] A dry 500 mL three-necked round-bottomed flask equipped with a magnetic stirring bar and an addition funnel, was charged with 65 g (0.40 mol) of 4-bromo-3-penten-2-one and 250 mL of anhydrous ether. To this mixture was added a solution of 45 g (0.44 mol) of triethylamine in 40 mL of ether dropwise at room temperature. After the mixture was stirred for 5 h at r.t., the precipitate of triethylamine hydrobromide was removed by suction and washed with ether (2×100 mL). The combined ether solution was washed with 5% hydrochloric acid. The solution was then washed with cold water (50 mL) and dried with magnesium sulfate. Evaporation and distillation afforded 21.5 g (65%) of 3,4-pentadien-2-one; bp 50 °C/60 mmHg.

## Biographical Sketches



**Shengming Ma** was born in Zhejiang Province (P. R. China) in 1965. He obtained his bachelor's degree at Hangzhou University in 1986. He then moved to Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences as a graduate student, and obtained his Ph.D. degree there in 1990 under the supervision of Professor Xiyan Lu. After working at the same institute as a Research Associate followed by an appointment as a Research Associate Professor, he joined Professor Venanzi's group at ETH (Switzerland) as a postdoctoral research fellow. One year later, he joined Professor Negishi's group at Purdue University. He returned to Shanghai Institute of Organic Chemistry in 1997. Currently he is a Research Professor and Director of State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences. His research interests are focused on development of new methodologies with an emphasis on OMCOS chemistry, especially asymmetric synthesis and their application to the highly selective synthesis of biologically active targets. He is the recipient of the Special Starting Grant for Outstanding Young Chemists issued by National Natural Science Foundation of China (29525202), 1999 Qiu Shi Award for Young Scientific Workers issued by Hong Kong Qiu Shi Foundation of Science and Technology (1999-2002), and the Award for Young Chemists issued by Chinese Chemical Society (1999).

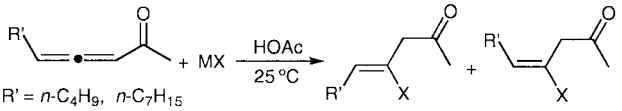


**Lintao Li** was born in Sichuan Province (P. R. China) in 1971. She received her bachelor's degree in 1994 from South-West Normal University. Under the supervision of Professor Guiyu Jin, she was awarded with her master degree in 1997 with an honor of "Excellent Thesis of Nankai University". She then moved to Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences and works there as a research associate under the supervision of Professor Shengming Ma. Her research interest is the development of new synthetic methodologies.

Downloaded by: University of Pittsburgh. Copyrighted material.

*Synthesis of 4-Iodo-4-penten-2-one*:[7] A 250 mL three-necked round-bottomed flask equipped with a magnetic stirring bar, an addition funnel, and a thermometer was charged with 24.3 g (0.16 mol) of lithium iodide, 23.4 g (0.39 mol) of acetic acid and 150 mL of tetrahydrofuran, and flushed with nitrogen. The flask was cooled with an ice-water bath and a solution of 10.7 g (0.13 mol) of 3,4-pentadien-2-one in 50 mL of tetrahydrofuran was added so that the temperature of the mixture did not rise above 0 °C. After stirring at 0 °C for ~8 h, the reaction mixture was evaporated, diluted with ether (250 mL), carefully neutralized with cool aqueous $K_2CO_3$. The two phases were separated and the aqueous layer was extracted with ether (2×100 mL). The combined organic layers were washed with saturated brine (3×100 mL) and dried over magnesium sulfate. Evaporation and distillation afforded 4-iodo-4-penten-2-one. Yield 16.9 g (62%), bp 62 °C/1 mmHg.

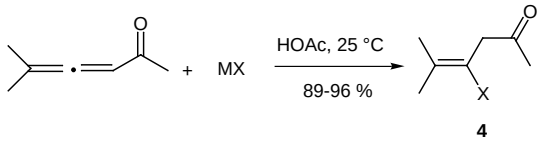
The hydroiodination and hydrobromination reactions of 3-alkyl-1,2-allenyl ketones with MX in HOAc at 25 °C yielded *Z*-isomer, i.e. *Z*-**3,** in high yield with fairly good stereoselectivity, although the corresponding hydrochlorination reaction afforded the products *E*-**3** and *Z*-**3** with lower stereoselectivities (Scheme 1). The formation of α,β-unsaturated β-haloenones was not observed.[7]




| MX | R' | NMR yield of *Z*-**3** (%) | NMR yield of *E*-**3** (%) |
|---|---|---|---|
| NaI | $n$-$C_4H_9$ | 90 | 3.6 |
| LiBr-$H_2O$ | $n$-$C_4H_9$ | 93 | 5 |
| LiCl-$H_2O$ | $n$-$C_4H_9$ | 25 | 55 |
| NaI | $n$-$C_7H_{15}$ | 87 | 3.5 |
| LiBr-$H_2O$ | $n$-$C_7H_{15}$ | 75 | 3.1 |
| LiCl-$H_2O$ | $n$-$C_7H_{15}$ | 18 | 54 |

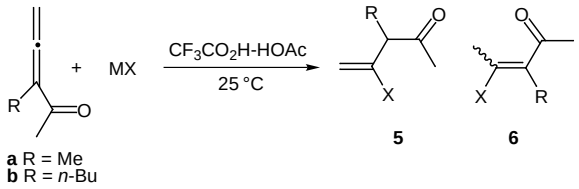
**Scheme 1**

Hydrohalogenation reaction of 5-methyl-3,4-hexadien-2-one with NaI, LiBr•$H_2O$, and LiCl•$H_2O$ in acetic acid at 25 °C also went smoothly to afford 4-halo-5-methyl-4-hexen-2-ones in high yields (equation 3). The β,γ carbon-carbon double bonds of **4** are stabilized by the four substituents, thus, the formation of α,β−unsaturated enones was not observed.[7]




**Equation 3**

Hydrohalogenation reaction of 3-methyl-3,4-pentadien-2-one with NaI in HOAc afforded a mixture of β,γ-unsaturated β-iodoenone **5a** and α,β-unsaturated β-iodoenone **6a** with an uninteresting ratio of 31:69. The corresponding reaction in $CF_3COOH$ afforded **5a** highly selectively, albeit in low yield (**5a**: **6a** = 96:4, 25%). Using a 3 equiv solution of $CF_3COOH$ in THF as the solvent greatly improved the yields, but the ratios of **5a**/**6a** still need optimization (78%, 80:20). Finally we found that the reaction could afford the β,γ-unsaturated β−iodoenone **5a** highly selectively when the reaction was carried out in a 1:1 mixture of $CF_3COOH$ and HOAc (82%, **5a**: **6a** = 96:4). Under these conditions, 3-methyl-3,4-pentadien-2-one or 3-butyl-3,4-pentadien-2-one reacted smoothly with NaI, LiBr-$H_2O$, and LiCl-$H_2O$ to afford the corresponding 4-halo-3-methyl(or butyl)-4-penten-2-ones **5** in high yields (70-99%) with high selectivity over the α,β-unsaturated enones **6** (2-5%) (equation 4).[7]




**Equation 4**

## 3 Hydrohalogenation Reaction of 1,2-Allenic Carboxylic Acids

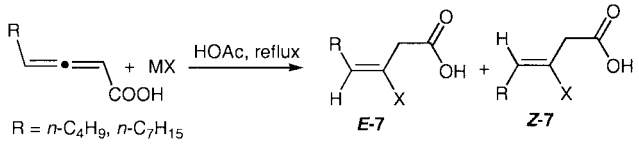
Reaction of 2,3-butadienoic acid with NaI, LiBr-$H_2O$, and LiCl-$H_2O$ in HOAc at 70 °C afforded the corresponding 3-halo-3-butadienoic acid with yields ranging from 87-92%.[6]

Hydroiodination reaction of 4-alkyl substituted 2,3-butadienoic acid with NaI in HOAc under reflux afforded *Z*-isomer of 3-halo-3-alkenoic acid with fairly good stereoselectivity. The corresponding reaction with LiBr•$H_2O$ or LiCl•$H_2O$ afforded a mixture of the products *Z*-**7** and *E*-**7** with low selectivities (Scheme 2). Products *Z*-**7** and *E*-**7** can be separated by column chromatography on silica gel.[9]

The hydrohalogenation reaction of 2-methyl-substituted 2,3-butadienoic acid with MX (M = Li, Na) in $CF_3CO_2H$ afforded the corresponding products in good yields (Scheme 3).[9]

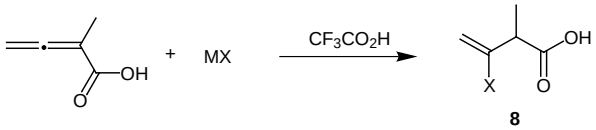


Downloaded by: University of Pittsburgh. Copyrighted material.




| R | MX | Yield (%) (*Z*/*E*)[a] |
|---|---|---|
| $n$-$C_4H_9$ | NaI | 95 (90/10) |
| $n$-$C_4H_9$ | LiBr-$H_2O$ | 89 (70/30) |
| $n$-$C_4H_9$ | LiCl-$H_2O$ | 87 (37/63) |
| $n$-$C_7H_{15}$ | NaI | 99 (80/20) |
| $n$-$C_7H_{15}$ | LiBr-$H_2O$ | 82 (64/36) |
| $n$-$C_7H_{15}$ | LiCl-$H_2O$ | 87 (43/37) |

**Scheme 2**



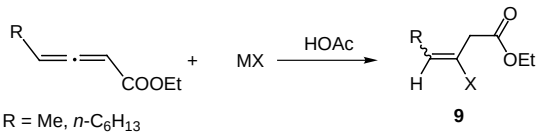

| MX | Temp. (°C) | Time (h) | Yield (%) |
|---|---|---|---|
| NaI | rt | 2.5 | 72 |
| LiBr·$H_2O$ | rt | 3 | 61 |

**Scheme 3**

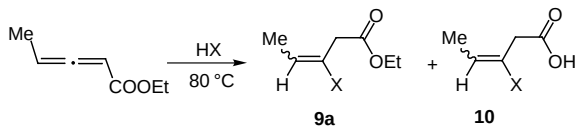
## 4 Hydrohalogenation Reaction of 1,2-Allenic Carboxylates

Hydrohalogenation reaction of unsubstituted methyl 2,3-butadienoate with MX in HOAc afforded methyl 3-halo-3-butenoates with yields ranging from 56-86%.[6]

The reaction of ethyl 4-alkyl substituted 2,3-alkadienoates with NaI, LiBr-$H_2O$, and LiCl-$H_2O$ in HOAc or $CF_3CO_2H$ went smoothly and afforded the corresponding ethyl 3-halo-3-alkenoates in high yields (70-98%), but the stereoselectivity is not high (equation 5).[9] However, a major side reaction, i.e., the hydrolysis of the ester group to carboxylic acid, was observed when ethyl 2,3-pentadienoate was treated with HX.[10] It afforded a mixture of ethyl 3-halo-3-pentenoate together with 3-halo-3-pentenoic acid (equation 6).[9]

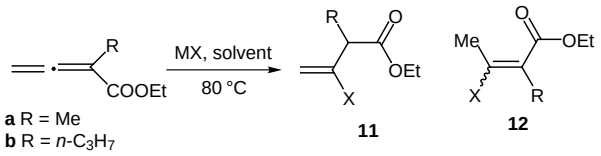



**Equation 5**



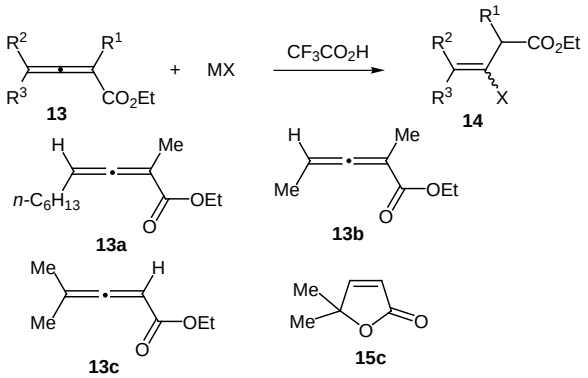

**Equation 6**

Hydrohalogenation reaction of ethyl 2-methyl-2,3-butadienoate with MX in HOAc afforded a mixture of 2-enoates **12a** and 3-enoates **11a** in low yields and selectivities (Yields of **12a** and **11a**: NaI: 10%, 41%; LiBr: <5%, <5%). However, when a 1:1 mixture of $CF_3CO_2H$ and HOAc or $CF_3CO_2H$ alone was used as the reaction medium, the hydrohalogenation went smoothly to afford ethyl 3-halo-2-methyl-3-butenoates **11a** in very good yields (80-95%) and the formation of 2-enoates **12a** was not observed. The corresponding reaction with HX was reported to afford a mixture of β,γ-unsaturated alkenoate and α,β-unsaturated alkenoate.[11] The reaction in $CF_3CO_2H$ was much more efficient than that in $CF_3CO_2H$-HOAc (1:1). Under the reaction conditions the hydrohalogenation reaction of ethyl 2-propyl-2,3-butadienoate also afforded ethyl 3-halo-2-propyl-3-butenoates in fairly good yields (63-98%) and the formation of 2-enoates **12b** was not observed (equation 7).[12]




**Equation 7**

For trisubstituted allenes, i.e., 2,3-dienoates **13a-b**, the reaction also went smoothly to afford the corresponding β,γ-unsaturated 3-enoates **14** (Scheme 4) with yields ranging from 55% to 89%.

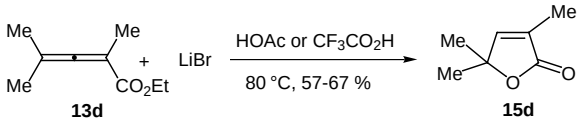



**Scheme 4**



Downloaded by: University of Pittsburgh. Copyrighted material.

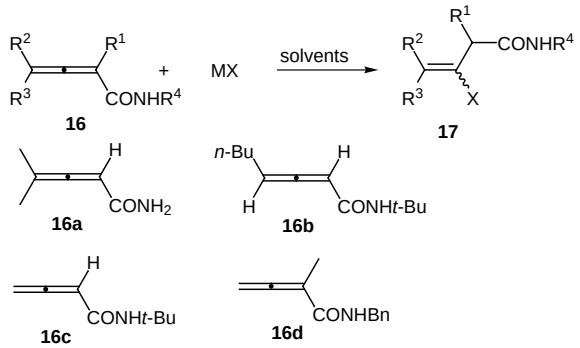
However, with 4,4-disubstituted 2,3-dienoate **13c**, cycloisomerization reaction product, i.e., butenolide **15c**,[13] was formed as the by-product along with the expected hydrohalogenation products **14**. When fully substituted 2,3-dienoate **13d** was treated with MX in HOAc, the cycloisomerization reaction affording butenolide **15d** was the only reaction observed (equation 8).[9]




**Equation 8**

## 5 Hydrohalogenation Reaction of 2,3-Dienamides

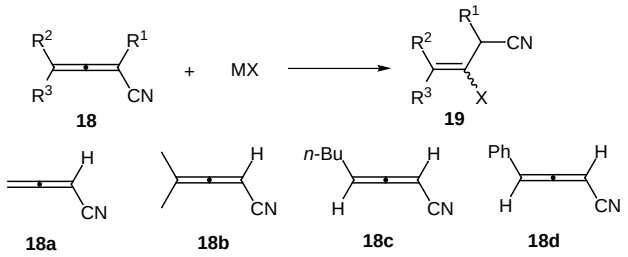
Hydrohalogenation reaction of 2,3-dienamide **16** usually afforded the corresponding 3-halo-3-alkenamides in high yields. Both MX-HOAc, MX-$CF_3CO_2H$ and aqueous HX systems can be applied. Hydrohalogenation reaction of 4-methyl-2,3-pentadienamide **16a** or 2,3-butadienamide **16c** with NaI, LiBr•$H_2O$, and LiCl•$H_2O$ in HOAc at 70 °C afforded the corresponding 3-halo-3-alkenamide **17** with yields ranging from 75% to 99%. Hydrohalogenation reaction of 2,3-octadienamide **16b** with MX in HOAc or $CF_3CO_2H$ at 70 °C afforded the corresponding 3-halo-3-octenamides **17** in high yields (69-98%) with low stereoselectivities. Both MX-$CF_3CO_2H$ and aqueous HX systems can be applied to *N*-benzyl 2-methyl-2,3-butadienamide (**16d**) (Scheme 5).[9]




**Scheme 5**

## 6 Hydrohalogenation Reaction of 2,3-Dienenitriles

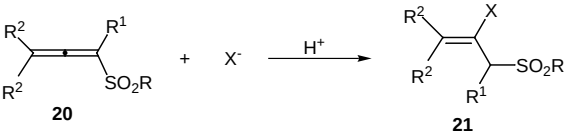
Hydrohalogenation of 2,3-butadienenitrile **18a** with MX in HOAc afforded the corresponding 3-halo-3-butenenitriles with yields ranging from 78% to 86%. However, the hydrohalogenation of 4-methyl-2,3-pentadienenitrile **18b** with MX in HOAc afforded the corresponding 3-halo-4-methyl-3-pentenenitriles in low yields (17-37%). When a 1:1 mixture of $CF_3CO_2H$ and HOAc was used as the reaction medium, the hydrohalogenation went smoothly to afford 3-halo-4-methyl-3-pentenenitriles in fairly good yields (76-80%) and excellent selectivities. The corresponding reactions with 2,3-octadienenitrile **18c** or 4-phenyl-2,3-butadienenitrile **18d** also went smoothly to afford 4-substitutied 3-halo-3-alkenenitriles **19** with yields ranging from 37% to 82% (Scheme 6), and 4-phenyl-2,3-butadienenitrile **18d** only afforded the *E*-isomer due to the existence of the phenyl group.




**Scheme 6**

## 7 Hydrohalogenation Reaction of 1, 2-Allenic Sulfones

Three sets of reaction conditions were established to carry out the hydrohalogenation reactions of 1,2-allenic sulfones **20** with metallic halides (Scheme 7).[14]




**20a**: R = $CH_3$, $R^1$ = H, $R^2$ = H
**20b**: R =Ph, $R^1$ = H, $R^2$ = H
**20c**: R = Ph, $R^1$ = $CH_3$, $R^2$ = H
**20d**: R = Ph, $R^1$ = H, $R^2$ = $CH_3$

Conditions A: MX (M = Na or Li), HOAc (solvent);
Conditions B: MX (M = Na or Li), HOAc (1 equiv), $CH_3CN$ (solvent);
Conditions C: HX (as the solvent and the reagent)

**Scheme 7**

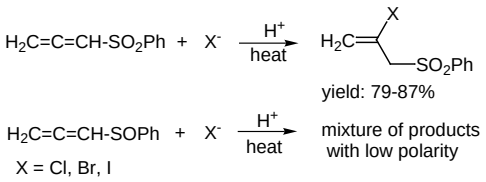
The reaction can proceed in HOAc (conditions A) or $CH_3CN$ using only 1 equiv. of HOAc (conditions B) with yields ranging from 67-80%, and the reaction of HX (conditions C) with 1,2-allenic sulfones also afforded the hydrohalogenation products in reasonable yields (65-83%). The migration of the C=C bond in product **21** was not observed even when **21** was heated in HOAc at 118 °C for 11 h.



Downloaded by: University of Pittsburgh. Copyrighted material.

## 8 Hydrohalogenation Reaction of 1, 2-Allenic Sulfoxides

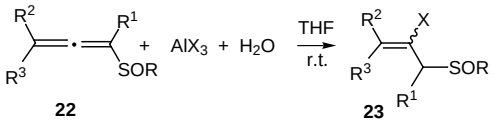
Hydrohalogenation reactions of 1,2-allenic sulfoxides with MX in HOAc failed to provide the expected 2-haloallylic sulfoxides (Scheme 8), probably due to the following two reasons: (1) the electron-withdrawing ability of sulfinyl group (RSO) is much weaker than that of sulfonyl group ($RSO_2$),[15] and (2) allenic sulfoxides are relatively unstable in acidic media such as HX at moderate temperature (50-70 °C).[16]




**Scheme 8**

After screening over numerous reaction conditions, $AlCl_3$ was found to show obvious activity, and a certain amount of water promoted this reaction. The ratio of $AlCl_3$ and $H_2O$ has big influence on the speed of this conversion and the reaction with 0.55 equiv. of $AlCl_3$ and 1 equiv. of water are the most efficient.[17]

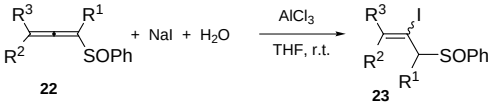
Under the reaction conditions, hydrohalogenation reactions of 1,2-allenic sulfoxides with $AlBr_3$ or $AlCl_3$ afforded the corresponding 2-haloallylic sulfoxides with yields ranging from 56% to 89%. The reaction with $AlBr_3$ was less efficient than the hydrochlorination reaction with $AlCl_3$. The structure of allenic sulfoxide is important: the reaction of some allenic sulfoxides, i.e. **22e**, **22f** and **22g,** did not proceed, probably due to the steric hindrance of the allenic sulfoxides. The hydrohalogenation yields for **22a** (81-89%) are usually higher than that for **22b** (64-71%), probably due to the higher stability of **22a** in the reaction media, since the control experiment showed that the reaction of **22b** was faster than that of **22a** (Scheme 9).[17]




**22a** R = Ph, $R^1 = R^2 = R^3 = H$;
**22b** R = $CH_3$, $R^1 = R^2 = R^3 = H$;
**22c** R = Ph, $R^1 = R^2 = H$, $R^3 = CH_3$;
**22d** R = Ph, $R^1 = R^2 = H$, $R^3 = n\text{-}C_4H_9$;
**22e** R = $CH_3$, $R^1 = CH_3$, $R^2 = R^3 = H$;
**22f** R = Ph, $R^1 = CH_3$, $R^2 = R^3 = H$;
**22g** R = Ph, $R^1 = H$, $R^2 = R^3 = CH_3$.

**Scheme 9**

Hydroiodination can be realized by using a combination of $AlCl_3$ and NaI (Scheme 10). When 4 equiv. of NaI and 0.55 equiv. of $AlCl_3$ were used, we did not observe the formation of any hydrochlorination product. However when we used 1 equiv. of **22a**, 1 equiv. of NaI, and 0.55 equiv. of $AlCl_3$, 20% of the hydrochlorination product was formed together with the hydroiodination product. The hydroiodination reaction of $PhS(O)CH{=}C{=}CHCH_3$ (**22c**) afforded the corresponding 2-iodoallyl sulfoxide in an isomeric *Z/E* ratio of 92: 8.[17]



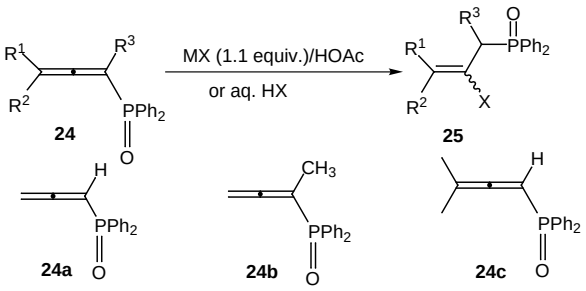

**22a** $R^1 = R^2 = R^3 = H$ yield: 79% (4 h)
**22c** $R^1 = R^2 = H$, $R^3 = CH_3$ yield: 69% (3 h); *Z/E* = 92/8

**Scheme 10**

The mechanism of this reaction is the Al(III)-mediated nucleophilic attack of $X^-$ at the center carbon atom of the electron-deficient allenes.

## 9 Hydrohalogenation Reaction of 1,2-Allenic Diphenylphosphine Oxides

Hydrohalogenation reactions of propadienyl diphenylphosphine oxide **24a** with MX (M = Na, Li, X = Cl, Br, I) in HOAc afforded 2-halo-2-propenyl diphenylphosphine oxides **25** in reasonable yields, ranging from 58% to 75%, while the similar reaction with aqueous HX afforded the products in much lower yields (31-46%). However, in cases of **24b** and **24c**, the reactivities towards MX in HOAc were low (Scheme 11).[9]




**Scheme 11**

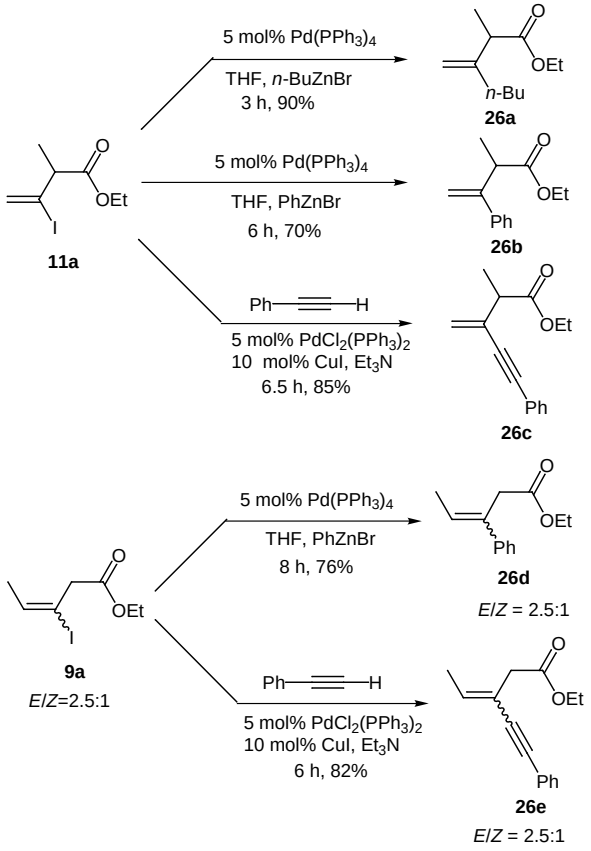
## 10 Synthetic Application

3-Halo-3-alkenoates are expected to be very useful synthetic intermediates, due to the existing carbon-halogen bond, ester group, carbon-carbon double bond, and the acidic α-proton. The carbon-iodine bond in ethyl 2-methyl-3-iodo-3-butenoate (**11a**) was shown to cross-couple



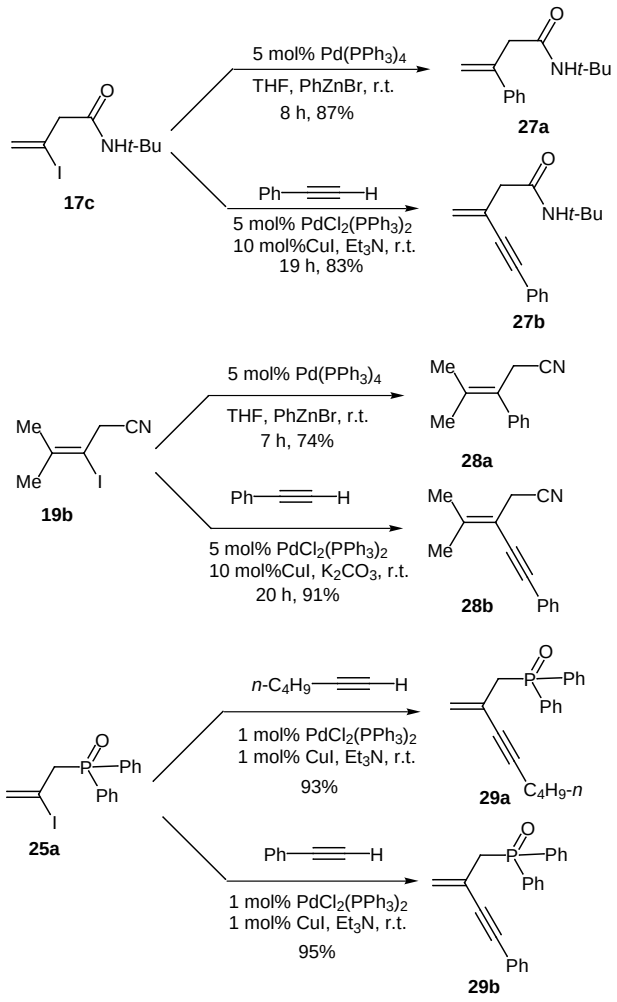
Downloaded by: University of Pittsburgh. Copyrighted material.

with terminal alkynes[18-20] and organozinc reagents[20,21] under the catalysis of Pd(0) to afford a series of substituted β,γ−unsaturated alkenoates (Scheme 12). Previously, **26b**-type of products were prepared via either the alkylation reaction of β-aryl-β-methyl acrylates[22] or dehydration reaction of β−hydroxycarboxylic acids which could be synthesized with a Reformatsky reaction,[23] while the synthesis of **26a**- and **26c**-type compounds have not been reported so far.




**Scheme 12**

The corresponding Sonogashira coupling and Negishi coupling reactions of **17c**, **19b**, and **25a** provided β-differently substituted 3-alkenamides, 3-alkenenitriles, and 2 differently substituted 2-propenyl diphenylphosphine oxides (Scheme 13).[9]




**Scheme 13**

## 11 Conclusion

We have developed a convenient methodology for the synthesis of 1-functionalized 2-halo-2-alkenes, such as 2-halo-2-alkenyl ketones, carboxylic acids, esters, amides, nitriles, sulfoxides (mediated by $AlX_3$), sulfones, and phosphine oxides via the simple reaction of electron-deficient allenes with MX in HOAc, $CF_3CO_2H$-HOAc, $CF_3CO_2H$ $CF_3CO_2H$ or THF. In most cases, the yields are high; the stereoselectivity depends on the reaction temperature and the solvent used; the ratios of β,γ-unsaturated and α,β-unsaturated products are determined by the acidity of solvent used. The uniquely located C=C bonds, C-X bonds, and the electron-withdrawing functional groups make these products synthetically useful. In most cases, the *Z*- and *E*-isomers can be readily separated providing efficient entries to both relevent stereoisomers.

## Acknowledgement

Financial support from Chinese Academy of Sciences, Shanghai Municipal Committee of Science and Technology, and the Major State Basic Research Development Program (Grant G2000077500) are greatly appreciated.

## References and Notes

(1) Schuster, H. F.; Coppola, G. M. *Allenes in Organic Synthesis*, John Wiley & Sons: New York, 1988. Patai, S. *The Chemistry of Ketenes, Allenes, and Related Compounds*, Part 1, John Wiley & Sons: New York, 1980.
(2) For some of our work in this area, see: Ma, S.; Shi, Z. *J. Org. Chem.* **1998**, *63*, 6387. Ma, S.; Shi, Z.; Yu, Z. *Tetrahedron*



Downloaded by: University of Pittsburgh. Copyrighted material.

*Lett.* **1999**, *40*, 2393. Ma, S.; Shi, Z.; Yu, Z. *Tetrahedron* **1999**, *55*, 12137. Ma, S.; Zhao, S. *J. Am. Chem. Soc.* **1999**, *121*, 7943. Ma, S.; Wu, S. *J. Org. Chem.* **1999**, *64*, 9314. Ma, S.; Zhang, J. *Chem. Commun.* **2000**, 117. Ma, S.; Li, L. *Org. Lett.* **2000**, *2*, 941. Ma, S.; Duan, D.; Shi, Z. *Org. Lett.* **2000**, *2*, 1419. Ma, S.; Zhao, S. *Org. Lett.* **2000**, *2*, 2495. Ma, S.; Xie, H. *Org. Lett.* **2000**, *2*, 3801. Ma, S.; Wei, Q.; Wang, H. *Org. Lett.* **2000**, *2*, 3893. Ma, S.; Gao, W. *Tetrahedron Lett.* **2000**, *41*, 8933. Ma, S.; Wu, S. *Chem. Commum.* **2001**, 441.

(3) For hydrohalogenation reaction of electron-deficient alkynes, see: Ma, S.; Lu, X. *J. Chem. Soc. Chem. Commun.* **1990,** 1643. Ma, S.; Lu, X. *Tetrahedron Lett.* **1990**, *31*, 7653. Ma, S.; Lu, X.; Li, Z. *J. Org. Chem.* **1992**, *57*, 709. Lu, X.; Zhu, G.; Ma, S. *Chin. J. Chem.* **1993**, *11*, 267. Ma, S.; Lu, X. *Org. Synth.* **1995**, *72*, 112. For an account, see: Lu, X.; Ma, S. *Chin. J. Chem.* **1998**, *16*, 388.

(4) Tidwell, T. T. *Org. React.* **1990**, *39*, 297 and references cited therein.

(5) Gras, J.-L.; Galledou, B. S. *Bull. Soc. Chim. Fr.* **1983**, II-89.

(6) Ma, S.; Shi, Z.; Li, L. *J. Org. Chem.* **1998**, *63*, 4522.

(7) Ma, S.; Li, L.; Xie, H. *J. Org. Chem.* **1999**, *64*, 5325.

(8) Buono, G. *Synthesis* **1981**, 872.

(9) Ma, S.; Xie, H.; Wang, G.; Zhang, J.; Shi, Z. *Synthesis* **2001**, 713.

(10) For the reaction of 2,3-allenoate with HX, see: Rosenthal, R. W.; Schwartzman, L. H. *J. Org. Chem.* **1959**, *24*, 836. Kurtz, P.; Gold, H.; Disselnkötter, H. *Ann.* **1959**, *624*, 1.

(11) Chalchat, J. C.; Duteurtre, P.; Théron, F.; Vessière, R. *C. R. Hebd. Seance Acad. Sc. Ser. C* **1971**, *273*, 832.

(12) Ma, S.; Wang, G. *Chin. J. Chem.* **1999**, *17*, 545.

(13) Marshall, A.; Wolf, M. A.; Wallace, E. M. *J. Org. Chem.* **1997**, *62*, 367.

(14) Ma, S.; Wei, Q. *J. Org. Chem.* **1999**, *64*, 1026.

(15) House, H. O. *Modern Synthetic Reaction*, 2nd, The Benjamin/ Cummings Publishing Company: Menlo Park, CA, 1972, p. 494.

(16) Barton, S. D.; Ollis, W. D. *Comprehensive Organic Chemistry*, Vol. 3, Pergamon Press: Oxford, 1979, p. 130.

(17) Ma, S.; Wei, Q. *Eur. J. Org. Chem.* **2000**, 1939.

(18) Sonogashira, K.; Tohda, Y.; Hagihara, N. *Tetrahedron Lett.* **1975**, 4467.

(19) Ma, S.; Shi, Z.; Yu, Z. *Tetrahedron Lett.* **1999**, *40*, 2393.

(20) Ma, S.; Shi, Z.; Yu, Z. *Tetrahedron* **1999**, *55*, 12137.

(21) For a seminal paper, see: Negishi, E.; King, A. O.; Okudado, N. *J. Org. Chem.* **1977**, *42*, 1821.

(22) Kano, S.; Yuasa, Y.; Yokomatsu, T.; Asami, K.; Shibuya, S. *J. Chem. Soc. Chem. Commun.* **1986**, 1717. Kano, S.; Yuasa, Y.; Asami, K.; Shibuya, S. *Heterocycles* **1988**, *27*, 1437.

(23) Theine, S. A.; Traynham, J. G. *J. Org. Chem.* **1974**, *39*, 153. Delaunay, J.; Orliac-Lemoing, A.; Moing, L.; Simonet, J. *Tetrahedron* **1988**, *44*, 7089.

Article Identifier:
1437-2096,E;2001,0,08,1206,1213,ftx,en;A26901ST.pdf

Downloaded by: University of Pittsburgh. Copyrighted material.